

Notes: Costinot, Vogel, and Wang (RES, 2013)
An Elementary Theory of Global Supply Chains

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1 Big picture

- **Question.** How does vertical specialization shape the interdependence of nations? The paper asks how countries located at different stages of a global supply chain are affected by global and local technological change.
- **Core idea.** Production is sequential and subject to mistakes. Because mistakes later in the chain destroy more accumulated value, more reliable countries have a comparative advantage in later stages.
- **Setting.** The model has many countries, one factor of production, a continuum of intermediate stages, and one final good. Each country differs in its probability of making mistakes along the chain.
- **Main equilibrium result.** In the unique free-trade equilibrium, countries vertically specialize: countries with higher failure rates produce earlier stages, while countries with lower failure rates produce later stages.
- **Why this matters.** Unlike a standard Ricardian model, absolute productivity differences become a source of comparative advantage because production is sequential.
- **Global technological change.** An increase in production complexity makes all countries move up the supply chain and increases inequality between nations. Standardization also makes all countries move up, but decreases inequality.
- **Local technological change.** Labor-augmenting progress and routinization in one country spill over to other countries through terms-of-trade effects. The direction and distributional consequences depend on where countries are located in the chain.
- **Extensions.** The paper shows how coordination costs, simultaneous production of multiple parts, imperfect observability of mistakes, and more general production functions can be incorporated without losing the central vertical-specialization logic.
- **Economic takeaway.** A country's position in a supply chain is not just descriptive. It determines which foreign technological changes help or hurt the country and how those changes affect international inequality.

2 Incremental contribution of the paper

- **A tractable theory of global supply chains.** Existing trade models with many countries, many goods, and sequential production are often difficult to solve. This paper gives a simple framework with sharp comparative statics.
- **Sequential production as comparative advantage.** The paper shows that absolute productivity differences can generate comparative advantage when stages must be performed in order.
- **Vertical specialization with many countries.** The model characterizes a unique free-trade equilibrium in which countries sort monotonically across stages of production.
- **Interdependence rather than only trade volumes.** The paper studies how technological changes in one country affect other countries in the same supply chain, not merely how fragmentation changes the volume of trade.

- **Global technology shocks.** The paper distinguishes complexity from standardization and shows that they have opposite implications for world inequality even though both push countries up the supply chain.
- **Local technology shocks.** The paper compares labor-augmenting progress and routinization, showing that local productivity improvements can have very different spillovers at the bottom and top of the chain.
- **A supply-side explanation for Linder-type patterns.** The model suggests why rich countries trade more with rich countries and trade higher-unit-value goods: they specialize in nearby, later stages of the same production chain.
- **A platform for extensions.** The paper keeps the baseline model simple, then demonstrates that the main sorting result survives several realistic features of global supply chains.

3 Model environment and key concepts

3.1 Countries, labor, and stages

- Countries are indexed by $c \in \{1, \dots, C\}$.
- Labor is the only primary factor. Country c has labor endowment L_c and wage w_c .
- Labor is perfectly mobile across stages within a country but immobile across countries.
- Production of the final good requires a continuum of stages $s \in (0, S]$.
- At each stage, producing the next intermediate good requires labor plus the intermediate good produced in the previous stage.

Interpretation: The model is designed to isolate the consequences of sequential production. The central question is not whether countries produce different goods, but where they locate within a production sequence.

3.2 Mistakes and failure rates

Each country has a failure rate $\lambda_c > 0$. Countries are ordered so that λ_c is decreasing in c , so higher-index countries are more reliable and more productive.

The basic production relation is:

$$q(s + ds) = (1 - \lambda_c ds)q(s).$$

- If a mistake occurs at stage s , the intermediate good is lost.
- Mistakes later in the chain are more costly because more value has already been embedded in the intermediate good.
- This creates a reason to assign more reliable countries to later stages.

Interpretation: A lower λ_c is valuable everywhere, but it is especially valuable late in the chain. This is the mechanism that turns absolute reliability differences into comparative advantage.

3.3 Prices, wages, and equilibrium objects

Let $p(s)$ be the world price of intermediate good s . The final good is the numeraire, so $p(S) = 1$. Profit maximization implies:

$$p(s + ds) \leq (1 + \lambda_c ds)p(s) + w_c ds,$$

with equality when country c actually produces stage $s + ds$.

The main equilibrium objects are:

- the sequence of cutoff stages $0 = S_0 < S_1 < \dots < S_C = S$;
- the measure of stages produced by country c , $N_c = S_c - S_{c-1}$;
- the export quantity Q_c of country c ;
- the wage vector $(w_1, \dots, w_C)$;
- export prices $p_c = p(S_c)$.

3.4 Vertical specialization

The central allocation result is:

$$Q_c(s) > 0 \quad \text{if and only if} \quad s \in (S_{c-1}, S_c].$$

- Country 1, the least reliable country, produces the earliest stages.
- Country C , the most reliable country, produces the latest stages.
- Each country produces one contiguous segment of the chain.

Economic intuition: In later stages, the intermediate input has high embedded value, so errors are especially costly. It is efficient and privately optimal to assign those stages to countries with lower failure rates.

4 Models and theoretical specifications

4.1 Free-trade equilibrium system

The pattern of vertical specialization and export levels satisfy:

$$S_c = S_{c-1} - \frac{1}{\lambda_c} \ln \left(1 - \frac{\lambda_c L_c}{Q_{c-1}} \right),$$

$$Q_c = e^{-\lambda_c(S_c - S_{c-1})} Q_{c-1}.$$

The wage and export-price system satisfies:

$$w_{c+1} = w_c + (\lambda_c - \lambda_{c+1})p_c,$$

$$p_c = e^{\lambda_c N_c} p_{c-1} + \left(e^{\lambda_c N_c} - 1 \right) \frac{w_c}{\lambda_c}.$$

- The first pair of equations uses labor-market clearing and material losses to determine stage assignments and export quantities.
- The second pair uses zero-profit conditions to determine relative wages and prices.
- The model is block-recursive: solve the allocation first, then solve wages and prices.

4.2 Global technological change: complexity

An increase in complexity is an increase in S , the total measure of stages needed to produce the final good.

- **Allocation effect.** All countries move up the supply chain.
- **Output effect.** A longer production chain increases cumulative losses, lowering total output for a given allocation.
- **Inequality effect.** Inequality between countries increases.

Economic intuition: More complex production raises the value of reliable production at later stages. Richer, more reliable countries capture a larger relative wage premium.

4.3 Global technological change: standardization

Standardization is modeled as a proportional decrease in all failure rates:

$$\lambda'_c = \beta \lambda_c, \quad \beta < 1.$$

- **Allocation effect.** All countries move up the supply chain.
- **Direct productivity effect.** All countries become more reliable.
- **Distributional effect.** Inequality between countries decreases.
- **Possible immiserizing growth.** The most technologically advanced country may lose because its terms of trade deteriorate.

Economic intuition: Standardization narrows the reliability gap between rich and poor countries. Poor countries gain disproportionately because they become more capable of performing later stages.

4.4 Local technological change: labor-augmenting progress

Labor-augmenting technical progress in country c_0 increases its effective labor endowment from L_{c_0} to $L'_{c_0} > L_{c_0}$.

- Countries below c_0 move down the supply chain.
- Countries above c_0 move up the supply chain.
- The country receiving the shock performs more stages.
- Inequality decreases among countries below or at c_0 .

- At the top of the chain, inequality can rise or fall because two forces conflict: moving up tends to increase inequality, while producing fewer stages tends to reduce it.

Interpretation: The spillover of a local labor-supply improvement is not monotone. The same shock can push some countries down and others up.

4.5 Local technological change: routinization

Routinization is a decrease in one country's failure rate λ_{c_0} , small enough not to change the ordering of countries.

- All countries move up the supply chain.
- Inequality rises among countries below c_0 .
- Inequality falls between c_0 and $c_0 + 1$ because country c_0 becomes closer in reliability to the country above it.
- Above c_0 , inequality is again non-monotone because moving up and producing fewer stages have opposing effects.

Interpretation: Routinization is different from labor-augmenting progress. It raises output by reducing losses, which creates additional labor demand above the improving country and pushes the whole chain upward.

4.6 Extensions

- **Coordination costs.** If crossing borders creates additional failure risk, vertical specialization still holds within each supply chain, but the structure of trade can change. Intermediate coordination costs can generate overshooting: more stages are offshored to a small country than under perfectly free trade.
- **Multiple parts and assembly.** With multiple sequential parts assembled into one final good, the poorest countries tend to specialize in assembly, while the richest countries specialize in later stages of the most complex parts.
- **Imperfect observability of mistakes.** If mistakes are not always observed, countries with better quality control may be valuable earlier in the chain because they prevent hidden defects from accumulating.
- **General production functions.** The main vertical-specialization result survives beyond Leontief production under conditions that keep labor cost shares lower at later stages and give more productive countries a relative advantage in later stages.

5 Identification and theoretical design

5.1 What the model is designed to identify

- The model identifies how sequential production alone can generate vertical specialization.
- It isolates terms-of-trade spillovers from global and local technological shocks.

- It shows how changes in a country's position along the supply chain translate into changes in relative wages.
- It provides qualitative comparative statics rather than quantitative estimates.

Interpretation: The paper is not an empirical identification exercise. Its contribution is theoretical identification of the mechanism linking sequential production, stage assignment, and international inequality.

5.2 Why the equilibrium predictions are sharp

- Countries differ in one key technological dimension: the failure rate λ_c .
- Stages are naturally ordered from early to late.
- Later stages embody more accumulated value, so failures are more costly later.
- This creates positive assortative matching between reliability and stage number.
- Labor-market clearing pins down how much of the chain each country performs.

Interpretation: The stark assumptions produce a very strong prediction: all countries occupy contiguous intervals, sorted by reliability.

5.3 Limitations of the theoretical design

- The baseline model has one factor of production, no capital, and homogeneous workers.
- It abstracts from firm boundaries, bargaining, market power, endogenous innovation, and policy.
- The baseline model has free trade and perfectly observable mistakes.
- The figures are simulations, not empirical estimates.
- The results are qualitative: they clarify mechanisms, but do not measure the magnitude of global supply-chain spillovers in the data.

Interpretation: The model's strength is tractability. Its weakness is that quantitative application requires additional structure.

6 Figures

6.1 Figure 1: Consequences of an increase in complexity

Read:

- Figure 1 shows a simulated increase in the length of the production process S .
- The left panel plots stage intervals. As complexity increases, countries move up the chain and perform different intervals of production.
- The right panel plots wages. Relative wage gaps widen as complexity rises.

- The figure visualizes Proposition 3: more complexity pushes all countries upward and increases inequality.

Economic message: More complex goods make reliability more valuable. Since more reliable countries are already at later stages, complexity raises their relative wages.

6.2 Figure 2: Consequences of standardization

Read:

- Figure 2 simulates a uniform decline in failure rates, parameterized by $1/\beta$.
- The left panel shows that all countries move up the chain.
- The right panel shows that relative wage inequality falls as standardization proceeds.
- The most advanced country can be hurt by terms-of-trade effects even though the technology improvement is global.

Economic message: Standardization narrows productivity gaps. It can help poorer countries move into later stages and reduce cross-country inequality.

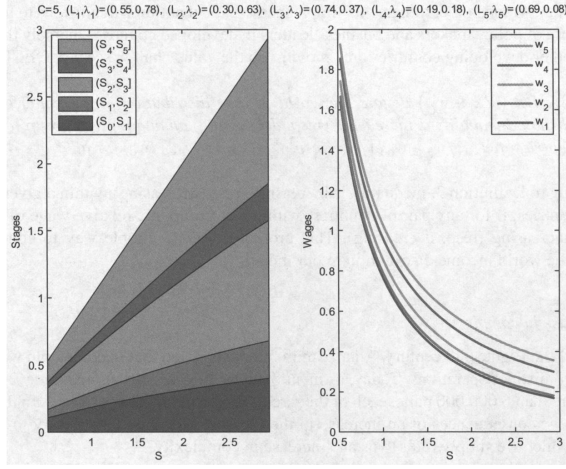


FIGURE 1
Consequences of an increase in complexity

Figure 1: Consequences of an increase in complexity.

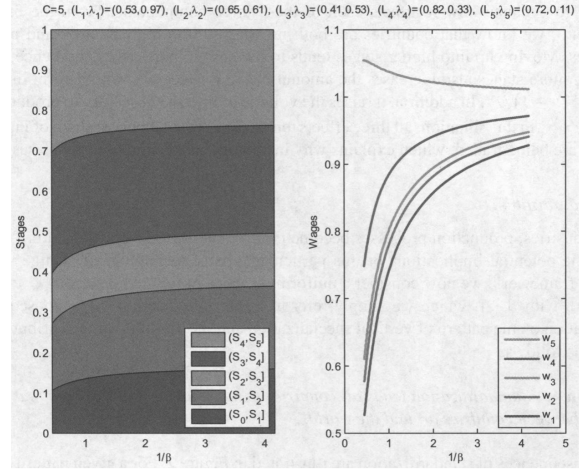


FIGURE 2

Figure 2: Consequences of standardization.

6.3 Figure 3: Consequences of labor-augmenting technical progress

Read:

- Figure 3 studies an increase in effective labor in country 3.
- The left panel shows that countries below the improving country move down, while countries above it move up.
- The right panel shows non-monotone wage responses.
- The figure visualizes Proposition 5: local labor-augmenting progress has asymmetric spillovers below and above the affected country.

Economic message: A local increase in effective labor reallocates production responsibilities across the chain. It can reduce inequality in some regions of the chain while increasing it in others.

6.4 Figure 4: Consequences of routinization

Read:

- Figure 4 studies a reduction in the failure rate of country 3.
- The left panel shows that all countries move up the supply chain.
- The right panel shows that wage inequality changes differently across parts of the chain.
- This contrasts with Figure 3: labor-augmenting progress and routinization are both local improvements, but they have different allocation effects.

Economic message: Not all productivity improvements are equivalent. Reducing mistakes changes the supply chain differently from expanding effective labor.

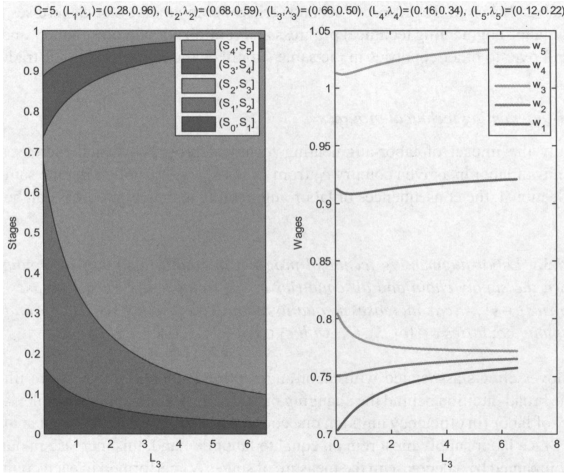


FIGURE 3
Consequences of labor-augmenting technical progress in country 3

Figure 3: Consequences of labor-augmenting technical progress in country 3.

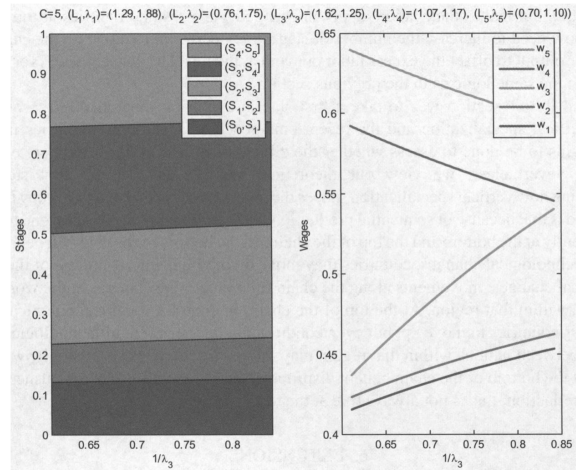


FIGURE 4

Figure 4: Consequences of routinization in country 3.

6.5 Figure 5: Consequences of a change in coordination costs

Read:

- Figure 5 studies the coordination-cost extension in a two-country case.
- The left panel shows how the set of stages performed by each country changes as coordination costs vary.
- The right panel shows the associated wage responses.
- The figure highlights overshooting: at intermediate coordination costs, the small or poorer country may perform more stages than under perfectly free trade.
- It also shows that the number of offshored stages is not a sufficient measure of interdependence.

Economic message: Lower coordination costs do not simply create a monotone expansion of offshoring. The structure of specialization can overshoot before converging to the free-trade allocation.

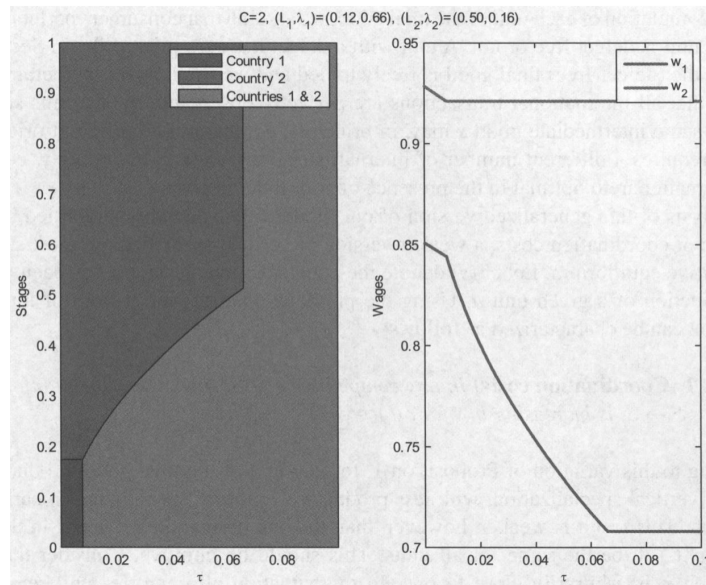


FIGURE 5

Figure 5: Consequences of a change in coordination costs.

7 Short summary

- The paper builds a simple theory of global supply chains with sequential production and mistakes.
- Countries differ in failure rates. More reliable countries specialize in later stages.
- The free-trade equilibrium has vertical specialization: countries occupy contiguous intervals of the chain.
- Complexity pushes all countries up the chain and increases inequality.
- Standardization also pushes all countries up the chain but decreases inequality and can hurt the most advanced country through terms-of-trade effects.
- Labor-augmenting progress in one country pushes countries below it down and countries above it up.
- Routinization in one country pushes all countries up.
- Spillovers differ sharply between the bottom and top of the chain.
- Coordination costs can generate overshooting, so the extent of offshoring is not always a good proxy for interdependence.
- The main takeaway is that the stage structure of production determines how countries share the gains and losses from technological change.

8 Conclusion

8.1 Core conclusion

Costinot, Vogel, and Wang develop a tractable theory in which production is sequential and mistakes can destroy intermediate goods. In this environment, more reliable countries specialize in later stages, and the position of each country in the chain determines how technological change affects wages, trade patterns, and international inequality.

8.2 Economic intuition

The key intuition is that later stages embody more prior work. A mistake late in production is therefore more costly than a mistake early in production. For that reason, reliability is especially valuable near the top of the chain. This creates a natural matching pattern: high-failure-rate countries produce early stages, and low-failure-rate countries produce later stages.

8.3 Incremental contribution revisited

The paper advances the theory of global supply chains by showing that a simple sequential-production model can generate strong results with many countries and a continuum of stages. It moves the literature beyond the question of whether production fragments across borders and asks how that fragmentation shapes the transmission of shocks across countries.

8.4 Implications

The model implies that trade in global supply chains creates deep interdependence. A technological improvement in one country can shift other countries' positions in the chain and change their wages. It also implies that policy discussions about "moving up the value chain" should distinguish between different forms of technological change, because labor-augmenting progress and routinization have different spillovers.

8.5 Limitations

The baseline model is intentionally stylized. It abstracts from capital, heterogeneous workers, firms, endogenous technology adoption, market power, strategic trade policy, and empirical magnitudes. The extensions show that the sorting logic is robust, but a quantitative evaluation of real-world supply chains would require additional data and structure.

8.6 Takeaway

The enduring takeaway is that global supply chains are not just collections of bilateral trade links. They are ordered production networks. Once production is sequential, a country's location in the chain becomes central for understanding who gains, who loses, and how technological change is transmitted across borders.